

A brief intro to SSI (Server Side Includes)

Background

SSI is a component of what's known as CGI (Common Gateway Interface) that can be used, with the help of web servers, to create some dynamic features. In this course we'll be concentrating on creating "common features" across an entire site, but I'll show you a couple of other cool stuff.

W' re not going through the ins and outs of CGI even at a basic level. Just remember a couple of buzzwords and do your job!

What you need to know

- Basic HTML (yes, that's right! H-T-M-L!)
- Whether your server is CGI-enabled.
- A bit of thinking and imagination

Does your web server support CGI?

If you use our Brunel web servers, you don't really need to do any tests, but you may want to do it any way to insure that you know how to do it in case you're on a different web server. A common example of a test code is displaying a server-side date.

1. Create a simple web page and include this piece of code exactly as you see here:

```
<!--#echo var="DATE_LOCAL" -->
somewhere within the body tags i.e.
<html>
<head>
<title>Test SSI </title>
</head>
<body>
<!--#echo var="DATE_LOCAL" -->
</body>
</html>
```

2. Save this file as **testssl.shtml (S - H - T - M - L not HTML)**
3. Put this page in your web space and check it out.

What have we learnt so far?

1. Any web page using this feature must use the unusual extension **shtml**. That is if your index page is going to use SSI, it needs to be renamed : *home.shtml* instead of *home.html*.
2. You must be very careful with spelling and casing (small/capital letters). Don't be like me!

Creating templates using SSI - A simple, but complete example

Let's say that you have a web site with three interlinked pages that share the same layout. One approach is that you create the layout page and save three times with three different name and then open each of the three pages individually, stick some contents into each and save them.

That's fine. What if you need to change/update the layout?

This is where SSI comes in. Now imagine that our web site has three pages: *index*, *images*, and *links*.

The common features of our layout consists of a Header (what's shown on the top) and a Footer (what's shown at the bottom of the page)

The final layout is going to look something like:

*Before we start create a directory in your *webhome* and call it *includes*

Step 1. Create the header:

Type the following text in your text editor and save it as *header.html*

```
<div id="header">
<h2> Stupid Web Design Studio</h2>
</div><!-- End of header -->
<!-- Nav starts -->
<div id="nav">
<ul>
    <li><a href="home.shtml">Home</a></li>
    <li><a href="folio.shtml">Folio</a></li>
    <li><a href="contact.shtml">Contact</a></li>
</ul>
<div class="clear"></div>
</div><!-- Nav ends -->
```

A bit unusual innit? The web page has no head, body, or html elements. Just do it!

Step 2. Create the footer:

Type the following text in your text editor and save it as *footer.html*

```
<div id="footer">
<p>All rights reversed. You may not visits this site before knowing the basics of XHTML</p>
</div><!-- End of footer -->
```

Step 3. Code the template page:

Type the code below in your text editor and save it as *template.shtml*

```
<html>
<head>
<title>Graduation Site</title>
<link rel="stylesheet" type="text/css" href="includes/styles.css" />
</head>
<body>
<!-- Header starts -->
<!--#include file="includes/header.html"-->
<!-- Header Ends -->
<div id="content">
<h2>Page Title</h2>
<p>
Vivamus fermentum orci sed lectus.
</p>
</div><!-- End of content -->
<!-- Footer starts -->
<!--#include file="includes/footer.html"-->
<!-- Footer ends -->
</body>
</html>
```

Step 4. Create the home, folio, and contact pages from the template

All we need to do now is open *template.shtml* and put whatever we want in the section between to comments:

```
<!-- contents code starts -->

<!-- contents code ends -->
```

and save each page respectively as: *home.shtml*, *folio.shtml*, and *contact.shtml*

That's if folks. Go ahead and make the other pages. If you want to see the web pages we just made go to: <http://people.brunel.ac.uk/~eesthnh/tmp/ssi> and inspect the code by viewing the page source. Anything unusual?

A must-do exercise

You're not going to see the power of SSI until you change some of the included files. So edit one of the files like footer.html, or header.html and see if the change appears across all pages.

Some fun with CGI

You need to place the codes shown here in a one of the shtml files. For instance, you can display the current date or time across the whole site by inserting a time code somewhere suitable in the *template.shtml*.

How to display the current date / time

Code

```
<!--#echo var="DATE_LOCAL" -->
<!--#config timefmt = "% d / % m / % y" -->
<!--#echo var="DATE_LOCAL" -->
<!--#config timefmt = "% H: % M: % S" -->
<!--#echo var="DATE_LOCAL" -->
```

Output sample

Wednesday, 12-Nov-2003 14:40:50 GMT

12/11/03

14: 46: 21

Other date and time formats can be generated using the following reference:

%a	abbreviated weekday name
%A	full weekday name
%b	abbreviated month name
%B	full month name
%c	locale's appropriate date and time
%C	default date and time format
%d	day of month - 01 to 31
%D	date as %m/%d/%y
%e	day of month - 1 to 31
%H	hour - 00 to 23
%I	hour - 01 to 12
%j	day of year - 001 to 366
%m	month of year - 01 to 12
%M	minute - 00 to 59
%n	insert a newline character
%p	string containing AM or PM
%r	time as %I:%M:%S %p
%R	time as %H:%M
%S	second - 00 to 59
%t	insert a tab character
%T	time as %H:%M:%S
%U	week number of year (Sunday is the first day of the week) - 00 to 53
%w	day of week - Sunday=0
%W	week number of year (Monday is the first day of the week) - 00 to 53
%x	Country-specific date format
%X	Country-specific time format
%y	year within century - 00 to 99
%Y	year as CCYY (4 digits)
%Z	timezone name

How to display the last modified date of a web page

Code

```
<!--#flastmod file="ssi.shtml" -->
```

Sample output

Wednesday, 12-Nov-2003 14:40:50 GMT

You can, of course, specify a format just before the above code and type "Last update" or something similar before `<!--#flastmod file="ssi.shtml" -->`.

So, what have we learnt so far?

('Nothing' is the honest answer, I know ;-)))

1. You can use SSI and some HTML to create a simple contents and layout management system.
2. The included page (header, footer, etc) are `<head>`less, `<body>`less, and `<html>`less.
3. All the files taking advantage of SSI have an shtml extension.
4. All you need is a bit of thinking and imagination.

A bit More Server-side includes with PHP

If you want to use PHP instead of SSI to include common sections of your website here's all you need to do:

1. Save your pages as .php . For example your home page or template will be home.php or template.php
2. Use PHP includes to include your common code for example:
3. Everything else is the same.

```
<html>
<head>
<title>Graduation Site</title>
<link rel="stylesheet" type="text/css" href="includes/styles.css" />
</head>
<body>
<!-- Header starts -->
<?php include 'includes/header.html'; ?>
<!-- Header Ends -->
<div id="content">
<h2>Page Title</h2>
<p>
Vivamus fermentum orci sed lectus.
</p>
</div><!-- End of content -->
<!-- Footer starts -->
<?php include 'includes/footer.html'; ?>
<!-- Footer ends -->
</body>
</html>
```

That's it!. Enjoy SSling :-)

Errors and omissions certainly expected.
Hooman Nili

Appendix

A – Screenshots of Site's Layout and pages



Message of the Day

- Stop moaning and learn the damn thing;
- Troubleshooting and thinking is what web design is about; it won't kill you;
- Everything has a fun part; try to find it;

Vivamus fermentum orci sed lectus. In hac habitasse platea dictumst. Integer mollis enim id ante. Nulla facilisi. Phasellus mattis est ut tortor. Phasellus id nunc. Maecenas id arcu. Curabitur ultricies, neque eget scelerisque posuere, metus mauris scelerisque turpis, at dictum elit nisi auctor libero

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News Flash

dfdsklfsfk sdfslf sdf ds kneweeds flet letctd

Update 08:-41-53

Folio

- Stop moaning and learn the damn thing;
- Troubleshooting and thinking is what web design is about; it won't kill you;
- Everything has a fun part; try to find it;

Vivamus fermentum orci sed lectus. In hac habitasse platea dictumst. Integer mollis enim id ante. Nulla facilisi. Phasellus mattis est ut tortor. Phasellus id nunc. Maecenas id arcu. Curabitur ultricies, neque eget scelerisque posuere, metus mauris scelerisque turpis, at dictum elit nisi auctor libero

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Contact

News Flash

dfdsklfsfk sdfslf sdf ds kneweeds flet letctd

Don't even think about contacting me

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Appendix

B – Site's CSS

```
body{
    font-family: arial,verdana;
    margin: 0;
    padding: 0;
}
.clear {
    clear: both;
}
#header {
    background: #e18c43;
    padding: 30px 20px 5px 20px;
    margin: 0;
    height: 80px;
}
#header h2 {
    margin: 0;
    color: #fff;
}
#nav {
    border-top: 2px solid #fff;
    background: #e1b040;
    height: 24px;
}
#nav ul {
    list-style: none;
    float: left;
    padding: 0;
    margin: 0;
    width: 500px;
}
#nav ul li {
    display: inline;
    float: left;
}
#nav ul a, #nav ul a:link, #nav ul a:visited {
    display: block;
    color: #fff;
    width: 80px;
    padding: 2px 5px;
    text-decoration: none;
    text-align: center;
}
#nav ul a:hover {
    background: #ffe381;
    color: #d35e4c;
}
#content {
    padding: 10px 20px;
    border-top: 2px solid #c9c9c9;
    border-bottom: 2px solid #c9c9c9;
}
#footer {
    background: #8193a8;
    padding: 1px 20px;
    color: #fff;
}
```